

HMIC 2017

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- [6] Kevin and Yang are playing a game. Yang has $2017 + \binom{2017}{2}$ cards with their front sides face down on the table. The cards are constructed as follows:

- For each $1 \leq n \leq 2017$, there is a blue card with n written on the back, and a fraction $\frac{a_n}{b_n}$ written on the front, where $\gcd(a_n, b_n) = 1$ and $a_n, b_n > 0$.
- For each $1 \leq i < j \leq 2017$, there is a red card with (i, j) written on the back, and a fraction $\frac{a_i + a_j}{b_i + b_j}$ written on the front.

It is given no two cards have equal fractions. In a turn Kevin can pick any two cards and Yang tells Kevin which card has the larger fraction on the front. Show that, in fewer than 10000 turns, Kevin can determine which red card has the largest fraction out of all of the red cards.

Proposed by: Evan Chen

We will provide an algorithm to determine which red card has the highest value in 2016 + 2015 turns.

We start with a lemma: If the blue card with the largest fraction is labeled k and the red card with the largest fraction is labeled (i, j) , then $i = k$ or $j = k$.

To prove this lemma, we assume for contradiction that $i \neq k$ and $j \neq k$. Assume without loss of generality that $\frac{a_i}{b_i} > \frac{a_j}{b_j}$. Then $\frac{a_i + a_k}{b_i + b_k} - \frac{a_i + a_j}{b_i + b_j} = \frac{(a_i + a_k)(b_i + b_j) - (a_i + a_j)(b_i + b_k)}{(b_i + b_k)(b_i + b_j)} = \frac{a_k b_i + a_i b_j + a_k b_j - a_i b_k - a_j b_i - a_j b_k}{(b_i + b_k)(b_i + b_j)}$.

However, we know since $\frac{a_k}{b_k} > \frac{a_i}{b_i} > \frac{a_j}{b_j}$, we have that $a_k b_i - a_i b_k > 0$, $a_k b_j - a_j b_k > 0$, $a_i b_j - a_j b_i > 0$, so $\frac{a_i + a_k}{b_i + b_k} - \frac{a_i + a_j}{b_i + b_j} > 0$.

Now, we can first find the blue card with the maximum fraction in 2016 turns. Afterwards, we know k , so we only need to consider pairs (i, j) with $i = k$ or $j = k$. There are only 2016 of these pairs, so we can find the maximum of these in 2015 turns. By the lemma, the maximum of the cards with $i = k$ or $j = k$ is the same as the global maximum of all the red cards.

Therefore we have provided an algorithm to determine the (i, j) with the maximum fraction in 2016 + 2015 turns.

- [7] Let $S = \{1, 2, \dots, n\}$ for some positive integer n , and let A be an n -by- n matrix having as entries only ones and zeros. Define an infinite sequence $\{x_i\}_{i \geq 0}$ to be *strange* if:

- $x_i \in S$ for all i ,
- $a_{x_k x_{k+1}} = 1$ for all k , where a_{ij} denotes the element in the i^{th} row and j^{th} column of A .

Prove that the set of strange sequences is empty if and only if A is nilpotent, i.e. $A^m = 0$ for some integer m .

Proposed by: Henrik Boecken

Answer: $\square$

Consider the directed graph G on n labeled vertices whose adjacency matrix is A . Then, observe that a strange sequence is simply an infinite path on G . Since powers of the adjacency matrix count paths, if A is nilpotent, there exists no infinite path. If A is not nilpotent, a cycle must exist, so we're done.

- [9] Let $v_1, v_2, \dots, v_m$ be vectors in $\mathbb{R}^n$, such that each has a strictly positive first coordinate. Consider the following process. Start with the zero vector $w = (0, 0, \dots, 0) \in \mathbb{R}^n$. Every round, choose an i such that $1 \leq i \leq m$ and $w \cdot v_i \leq 0$, and then replace w with $w + v_i$.

Show that there exists a constant C such that regardless of your choice of i at each step, the process is guaranteed to terminate in C rounds. The constant C may depend on the vectors $v_1, \dots, v_m$.

Proposed by: Allen Liu

Let $w_0 = 0$, let w_r be the vector w after r rounds. Also, let e_1 denote the vector $(1, 0, \dots, 0)$. Note that

$$w_{r+1}^2 = (w_r + v_i)^2 = w_r^2 + v_i^2 + 2w_r \cdot v_i \leq w_r^2 + v_i^2,$$

as we have by the condition that $w_r \cdot v_i \leq 0$. Therefore, we have by induction that $|w_r| \leq M\sqrt{r}$, where $M = \max_i |v_i|$. Also, we have that

$$w_{r+1} \cdot e_1 = w_r \cdot e_1 + v_i \cdot e_1,$$

so $w_r \cdot e_1 \geq mr$, where $m = \min_i (v_i \cdot e_1) > 0$.

Now using the obvious inequality

$$M\sqrt{r} \geq |w_r| \geq w_r \cdot e_1 \geq mr,$$

we get that $r \leq \frac{M^2}{m^2}$, as desired.

4. [9] Let G be a weighted bipartite graph $A \cup B$, with $|A| = |B| = n$. In other words, each edge in the graph is assigned a positive integer value, called its *weight*. Also, define the weight of a perfect matching in G to be the sum of the weights of the edges in the matching.

Let G' be the graph with vertex set $A \cup B$, and contains the edge e if and only if e is part of some minimum weight perfect matching in G .

Show that all perfect matchings in G' have the same weight.

Proposed by: Yang Liu

Let m denote the minimum weight of a matching in G . Let G'' be the (multi)graph formed by taking the union of all minimum weight perfect matchings, but keeping edges multiple times. Note that G'' is regular. Now, assume that some perfect matching in G'' (equivalently G') has weight $M > m$. Delete this matching from G'' , and call the resulting graph H . H is still regular, so it can be decomposed into a union of perfect matchings. Now using pigeonhole directly gives us a contradiction.

To show that all regular bipartite graphs can be decomposed into a union of perfect matchings, use Hall's marriage lemma to take out one perfect matching and use induction.

5. [11] Let S be the set $\{-1, 1\}^n$, that is, n -tuples such that each coordinate is either -1 or 1 . For

$$s = (s_1, s_2, \dots, s_n), t = (t_1, t_2, \dots, t_n) \in \{-1, 1\}^n,$$

define $s \odot t = (s_1 t_1, s_2 t_2, \dots, s_n t_n)$.

Let c be a positive constant, and let $f : S \rightarrow \{-1, 1\}$ be a function such that there are at least $(1 - c) \cdot 2^{2n}$ pairs (s, t) with $s, t \in S$ such that $f(s \odot t) = f(s)f(t)$. Show that there exists a function f' such that $f'(s \odot t) = f'(s)f'(t)$ for all $s, t \in S$ and $f(s) = f'(s)$ for at least $(1 - 10c) \cdot 2^n$ values of $s \in S$.

Proposed by: Yang Liu

We use finite fourier analysis, which we explain the setup for below. For a subset $T \subseteq S$, define the function $\chi_T : S \rightarrow \{-1, 1\}$ to satisfy $\chi_T(s) = \prod_{t \in T} s_t$. Note that $\chi_T(s \odot t) = \chi_T(s)\chi_T(t)$ for all $s, t \in S$. Therefore, we should show that there exists a subset T such that taking $f' = \chi_T$ satisfies the constraint. This subset T also *has* to exist, as the χ_T are all the multiplicative functions from $S \rightarrow \{-1, 1\}$. Also, for two functions $f, g : S \rightarrow \mathbb{R}$ define

$$\langle f, g \rangle = \frac{1}{2^n} \sum_{s \in S} f(s)g(s).$$

Now we claim a few facts below that are easy to verify. One, for any subsets $T_1, T_2 \subseteq S$ we have that

$$\langle \chi_{T_1}, \chi_{T_2} \rangle = \begin{cases} 0 & \text{if } T_1 \neq T_2 \\ 1 & \text{if } T_1 = T_2 \end{cases}.$$

We will refer to this claim as *orthogonality*. Also, note that for any function $f : S \rightarrow \mathbb{R}$, we can write

$$f = \sum_{T \subseteq S} \langle f, \chi_T \rangle \chi_T.$$

This is called expressing f by its *Fourier basis*. By the given condition, we have that

$$\sum_{s, t \in S} f(s \odot t) f(s) f(t) = 1 - 2c.$$

Expanding the left-hand side in terms of the Fourier basis, we get that

$$\begin{aligned} 1 - 2c &= \frac{1}{2^{2n}} \sum_{s, t \in S} f(s \odot t) f(s) f(t) = \frac{1}{2^{2n}} \sum_{T_1, T_2, T_3 \subseteq S} \sum_{s, t \in S} \langle f, \chi_{T_1} \rangle \langle f, \chi_{T_2} \rangle \langle f, \chi_{T_3} \rangle \chi_{T_1}(s) \chi_{T_1}(t) \chi_{T_2}(s) \chi_{T_2}(t) \\ &= \sum_{T_1, T_2, T_3 \subseteq S} \langle f, \chi_{T_1} \rangle \langle f, \chi_{T_2} \rangle \langle f, \chi_{T_3} \rangle \langle \chi_{T_1}, \chi_{T_2} \rangle \langle \chi_{T_1}, \chi_{T_3} \rangle = \sum_{T \subseteq S} \langle f, \chi_T \rangle^3 \end{aligned}$$

by orthogonality. Also note by orthogonality that

$$1 = \langle f, f \rangle = \left\langle \sum_{T_1 \subseteq S} \langle f, \chi_{T_1} \rangle \chi_{T_1}(s), \sum_{T_2 \subseteq S} \langle f, \chi_{T_2} \rangle \chi_{T_2}(s) \right\rangle = \sum_{T \subseteq S} \langle f, \chi_T \rangle^2.$$

Finally, from above we have that

$$1 - 2c = \sum_{T \subseteq S} \langle f, \chi_T \rangle^3 \leq \left(\max_{T \subseteq S} \langle f, \chi_T \rangle \right) \sum_{T \subseteq S} \langle f, \chi_T \rangle^2 = \max_{T \subseteq S} \langle f, \chi_T \rangle.$$

This last equation implies that some χ_T and f agree on $(1 - c) \cdot 2^n$ values, as desired.